

RAHUL VERMA

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SUMMARY

Computer Science Engineering student with hands-on experience in backend development, full-stack applications, and cross-platform mobile solutions. Skilled in designing RESTful APIs, database-driven systems, and scalable software applications using modern development technologies. Experienced in building real-world solutions integrating AI capabilities, intelligent workflows, and real-time communication systems.

SKILLS

Programming Languages: Python, JavaScript, TypeScript, C++, SQL

Frontend Development: HTML, CSS, React.js, React Native, Tailwind CSS

Backend Development: Node.js, Express.js, Flask, REST APIs, FCM, JWT Authentication

Database Management: Microsoft SQL Server, MySQL, MongoDB, PostgreSQL, DynamoDB

AI & Machine Learning: Retrieval-Augmented Generation (RAG), Vector Embeddings, LLM Integration

Tools & Platforms: Git, GitHub, Linux, Postman, AWS, Render, Clever Cloud, Google APIs

PROJECT

AttendEase – Academic Scheduling & Institutional Workflow Automation

AttendEase is a production-ready institutional system designed to automate timetable scheduling, class cancellations, substitutions, and other communication workflows. It provides a centralized platform for real-time academic operations.

- Engineered robust **REST APIs** for scheduling, authentication, leave management, substitutions, and announcements to enable instantaneous updates and automated event notifications.
- Implemented **Role-Based Access Control (RBAC)** for secure multi-role access. Designed a SQL Server database with normalization and strong relational integrity to prevent conflicts.
- Integrated a React Native mobile application to ensure seamless cross-platform accessibility and dynamic, on-the-fly data synchronization.
- Implemented **BLE Mesh-based** offline notification sharing to improve system reliability.

AI Larvix – Branching Ideas into Trees for Deep Research

AI Larvix is an AI research system designed to **structure ideas into branching trees**, enabling **deep exploration of topics** through organized multi-path reasoning.

- Architected a topic-level branching mechanism enabling divergence of conversations into independent **subtrees** while preserving parent lineage.
- Enabled systematic deep research exploration, allowing parallel analysis of subtopics without cross-topic interference.
- Improved research efficiency and scalability by reducing context contamination, **lowering token overhead**, and accelerating inference in long-form exploratory sessions.

V2Vision – Vehicle-to-Vehicle Communication System

V2Vision is a high-performance live vehicle communication and tracking platform designed to enhance situational awareness through GPS-based positioning and server-driven vehicle-to-vehicle (V2V) data exchange.

- Engineered a dynamic interface to stream **live** telemetry data—including position, velocity, heading, and vehicle classification—for all surrounding traffic within a 300–500m radius.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science Engineering

Dr. A.P.J. Abdul Kalam Government Institute of Technology • Tanakpur, Uttarakhand • 2026

Senior Secondary (Class 12), CBSE

Kendriya Vidyalaya No. 2, NHPC Banbasa • Uttarakhand • 2022

Secondary (Class 10), CBSE

Kendriya Vidyalaya No. 2, NHPC Banbasa • Uttarakhand • 2020